

# Arrhythmias — Idiopathic VT (RVOT & Fascicular)





## 1. Idiopathic VT = normal heart, benign

### WHAT YOU SEE

Sign 'IDIOPATHIC VT — NORMAL HEART / NO SCAR / BENIGN'

### WHAT IT ENCODES

Idiopathic VT occurs in a structurally normal heart — no scar, normal EF, normal coronaries, normal imaging — and carries a benign prognosis with very low SCD risk. It is the exception to the rule that VT means structural disease. Diagnosis is one of exclusion: normal echo, normal MRI (no late gadolinium enhancement), and exclusion of ischemia, ARVC, and channelopathies.

### BOARD TRAP

WRONG: assuming all sustained VT implies structural heart disease and an ICD — idiopathic VT in a normal heart is benign and managed with drugs/ablation, not an ICD. Confirm a truly normal heart before applying this benign label.

## 2. No ICD needed

### WHAT YOU SEE

'ICD NEEDED' crossed out

### WHAT IT ENCODES

Because idiopathic VT arises in a structurally normal heart with negligible SCD risk, an ICD is generally NOT indicated — treatment is symptom-directed with beta-blockers/CCBs and curative catheter ablation. Distinguishing idiopathic VT from scar/structural VT (which DOES warrant an ICD) is the management-defining step. The discriminator is presence vs absence of structural disease, not the VT itself.

### BOARD TRAP

WRONG: implanting an ICD for idiopathic RVOT VT — it is benign and curable by ablation, so an ICD is unnecessary. The trap reverses the rule: ICDs are for scar/structural VT and inherited high-risk conditions, not idiopathic VT in a normal heart.

## 3. RVOT VT — most common idiopathic

### WHAT YOU SEE

'RVOT — MOST COMMON IDIOPATHIC'

### WHAT IT ENCODES

Right ventricular outflow tract VT is the most common idiopathic VT, arising from cAMP-mediated triggered activity just below the pulmonic valve. Classic presentation: a young/middle-aged patient with palpitations or exercise/stress-induced VT, often as repetitive nonsustained runs (monomorphic) with a structurally normal heart. It is frequently catecholamine-sensitive and adenosine-sensitive.

### BOARD TRAP

WRONG: equating LBBB-pattern outflow VT with a malignant disease — RVOT VT is benign, BUT a similar LBBB/inferior-axis VT can be the first sign of ARVC, so structural disease must be excluded before calling it RVOT VT.

## 4. LBBB + inferior axis

### WHAT YOU SEE

'LBBB + INFERIOR AXIS'

### WHAT IT ENCODES

RVOT VT shows an LBBB morphology (dominant S in V1, originating in the right ventricle) with an INFERIOR axis (tall positive R waves in II, III, aVF) because the focus sits high in the outflow tract and depolarizes downward. Recognizing 'LBBB + inferior axis, normal heart, young patient' is the board signature of RVOT VT.

### BOARD TRAP

WRONG: reading LBBB-morphology VT as proof of a left-sided/benign origin — LBBB pattern actually indicates a RIGHT ventricular (or septal) exit. The morphology localizes the exit site; pair it with the inferior axis to identify RVOT.

## 5. Triggered automaticity

### WHAT YOU SEE

'TRIGGERED AUTOMATICITY'

### WHAT IT ENCODES

The mechanism of RVOT VT is cAMP-mediated triggered activity (delayed afterdepolarizations from calcium overload), making it catecholamine-sensitive and provoked by exercise, stress, or caffeine. This explains its response to beta-blockers, calcium-channel blockers, adenosine, and vagal maneuvers — therapies that lower cAMP/calcium.

### BOARD TRAP

WRONG: assuming RVOT VT is scar-based reentry like post-MI VT — it is triggered activity, which is why it responds to beta-blockers/adenosine. Reentrant scar VT does not terminate with adenosine.

## 6. Beta-blocker / CCB first-line

### WHAT YOU SEE

Ladder 'BB / CCB' first-line, 'FLECAINIDE'

### WHAT IT ENCODES

First-line drug therapy for symptomatic RVOT VT is a beta-blocker or a non-dihydropyridine calcium-channel blocker (verapamil/diltiazem), targeting the catecholamine/cAMP-driven mechanism. Flecainide is a reasonable option BECAUSE the heart is structurally normal. Adenosine or vagal maneuvers can terminate acute episodes.

### BOARD TRAP

WRONG: giving flecainide (or other class Ic agents) to a VT patient WITH structural/ischemic heart disease — class Ic drugs are contraindicated there (CAST trial, proarrhythmic). Flecainide is acceptable ONLY because idiopathic VT occurs in a structurally normal heart.

## 7. Ablation ~80% curative

### WHAT YOU SEE

'ABLATION 80%'

### WHAT IT ENCODES

Catheter ablation of the focal RVOT origin is highly effective (~80–90% cure) and is the definitive therapy for symptomatic, frequent, or drug-refractory idiopathic RVOT VT. Because the heart is normal and the focus is discrete, ablation is low-risk and often offered early, even as a first-line choice in motivated patients.

### BOARD TRAP

WRONG: reserving ablation only for drug failures in RVOT VT — given its high cure rate and low risk in a normal heart, ablation is a reasonable first-line option, not merely a last resort.



## 8. Fascicular VT (Belhassen)

### WHAT YOU SEE

Sign 'FASCICULAR VT — BELHASSEN / PURKINJE REENTRY'

### WHAT IT ENCODES

Fascicular (Belhassen) VT is the second main idiopathic VT — a REENTRANT circuit within the left posterior fascicle / Purkinje network of the left ventricle (mechanism differs from RVOT's triggered activity). It typically affects young men, is often relatively narrow for a VT, and is the classic verapamil-sensitive VT.

### BOARD TRAP

WRONG: lumping fascicular VT with RVOT mechanistically — RVOT is triggered automaticity (beta-blocker/adenosine sensitive) while fascicular VT is reentry (verapamil sensitive). Mechanism dictates the drug that works.

## 9. RBBB + left axis

### WHAT YOU SEE

'RBBB + LAD'

### WHAT IT ENCODES

Fascicular VT shows an RBBB morphology (dominant R in V1, left-ventricular origin) with LEFT-axis deviation, reflecting its exit from the left posterior fascicle. (A less common left-anterior-fascicular variant gives RBBB with right-axis deviation.) The RBBB + left-axis pattern in a young patient is the fascicular VT signature.

### BOARD TRAP

WRONG: confusing the RBBB+left-axis fascicular pattern with the RVOT LBBB+inferior-axis pattern — they are mirror clues. RBBB/left-axis → fascicular (verapamil); LBBB/inferior-axis → RVOT (beta-blocker/CCB).

## 10. IV verapamil terminates

### WHAT YOU SEE

Cart 'IV VERAPAMIL TERMINATES — calci-yum'

### WHAT IT ENCODES

Fascicular (Belhassen) VT is uniquely verapamil-sensitive: IV verapamil (or diltiazem) terminates it because the reentry circuit depends on calcium-dependent slow-conducting Purkinje tissue. Mnemonic: 'calci-yum' — fascicular VT loves calcium-channel blockers. Catheter ablation is curative for recurrent cases.

### BOARD TRAP

WRONG: giving verapamil to ANY wide-complex tachycardia — in scar-related VT (structural heart disease) IV verapamil causes profound hypotension and hemodynamic collapse. Verapamil is appropriate ONLY for confirmed idiopathic fascicular VT in a normal heart.

## 11. Two most common idiopathic VTs

### WHAT YOU SEE

Sign 'TWO MOST COMMON IDIOPATHIC VTs' (RVOT, fascicular)

### WHAT IT ENCODES

The two most commonly tested idiopathic VTs are RVOT VT (most common overall; triggered activity; LBBB+inferior axis; beta-blocker/CCB) and fascicular/Belhassen VT (Purkinje reentry; RBBB+left axis; verapamil). Knowing the paired morphology and drug for each is the high-yield board takeaway. Both occur in structurally normal hearts.

### BOARD TRAP

WRONG: mismatching the drug to the VT — RVOT responds to beta-blockers/adenosine; fascicular to verapamil. Swapping them (e.g., verapamil for RVOT 'because both are idiopathic') is the common error.

## 12. Exclude ARVC

### WHAT YOU SEE

Sign 'EXCLUDE ARVC!'

### WHAT IT ENCODES

Before labeling outflow-tract VT 'idiopathic,' exclude arrhythmogenic right ventricular cardiomyopathy (ARVC) — a malignant desmosomal disease that also produces LBBB-pattern VT but causes SCD in young athletes. Red flags: epsilon waves and T-wave inversions in V1–V3 on ECG, RV dilation/dysfunction with fibrofatty replacement on MRI, and family history of SCD. ARVC needs an ICD; idiopathic RVOT VT does not.

### BOARD TRAP

WRONG: calling LBBB-morphology VT automatically benign RVOT VT — ARVC mimics it but is malignant. The discriminator is structural imaging and ECG repolarization/depolarization abnormalities (T-wave inversions V1–V3, epsilon wave); a normal MRI is required to call it idiopathic.

## 13. CPVT — bidirectional/exercise VT

### WHAT YOU SEE

Greenhouse panel 'CPVT / BIDIRECTIONAL VT' with exercise/short-QT motifs

### WHAT IT ENCODES

Catecholaminergic polymorphic VT (CPVT) is an inherited RyR2 channelopathy (calcium leak → delayed afterdepolarizations) causing exercise- or emotion-induced bidirectional or polymorphic VT and syncope/SCD in young people, with a structurally normal heart and normal resting ECG. Treatment: high-dose beta-blockers (nadolol), flecainide, exercise restriction, and ICD for high-risk/aborted arrest. Bidirectional VT also classically suggests digoxin toxicity.

### BOARD TRAP

WRONG: dismissing exertional syncope with a normal resting ECG as benign or RVOT VT — CPVT has a normal baseline ECG and only reveals itself on exercise/stress testing. Bidirectional VT should trigger thought of CPVT or digoxin toxicity, not idiopathic RVOT VT.

## 14. Tachycardia-induced cardiomyopathy

### WHAT YOU SEE

Heart-shaped tree 'TACHYCARDIA-INDUCED CMP' marked 'REVERSIBLE'

### WHAT IT ENCODES

Incessant or very frequent idiopathic VT/PVCs can cause a tachycardia-induced (tachycardiomyopathy) reduction in LV function that is REVERSIBLE — controlling or ablating the arrhythmia restores the ejection fraction. So a low EF in this setting may be the CONSEQUENCE of the arrhythmia, not evidence of primary structural disease. Suspect it with high PVC burden (often >10–15%) and otherwise unexplained cardiomyopathy.

### BOARD TRAP

WRONG: attributing a low EF in a patient with incessant idiopathic VT to irreversible primary cardiomyopathy and writing off recovery — the LV dysfunction is often reversible after arrhythmia control/ablation. Don't mistake tachycardia-induced cardiomyopathy for a fixed structural disease requiring lifelong ICD.